

Positive Bath Filtering and Complete-Word Composition for a Conditional MQM Latch

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Abstract—We construct a positive finite-window filter-Gram generator for the previously declared Ohmic reservoir and separately compose a terminal latch with the complete thirteen-step execute–capture–renew word. Five frozen coarse-graining choices preserve positivity but fail the port accuracy target at the original deadline. An explicit generator-comparison inequality remains inconclusive and supplies no microscopic bath certificate. In the separate local-bath history model, the complete bank word has a directed interval certificate below 2^{-16} around a later sampled deadline, retaining output and donor correlations. These results distinguish valid generator construction, conditional operational correctness and physical reservoir validity; they do not establish a qubit two-local implementation or quantum advantage.

Index Terms—Positive coarse graining, history Hamiltonians, complete-domain accounting, quantum reservoirs, MQM.

I. INDEPENDENT FROZEN WORKLOADS

The previous terminal marker uses a native ququart, a completion flag and an auxiliary qubit. Its admitted sector has dimension eight, and its local-bath certificate assumes an initially prepared port state, identity coupling to stored data and a traced Markov bath. Here two distinct workloads address remaining gaps: finite-window bath filtering on that port, and composition of the full body word with a local-bath terminal marker. Their certificates are not interchangeable.

For bath filtering, retain the laboratory scale $\Omega = 64$, cutoff $\omega_c = 640$, $\beta\Omega = 20$, reference downward rate 8 and coupling observable $I \otimes X_R$. Freeze coarse-graining times $\tau \in \{1/64, 1/16, 1/4, 1, 4\}$, target 2^{-16} and port readout $t = 16$. No material calibration or microscopic continuum reference is available. Lamb shifts are omitted and assumed compensated, with that debit still OPEN.

For word composition, freeze uniform program-edge hopping one, local damping eight, terminal exchange $\lambda^2 = 4\sqrt{2}$, times 16, 64, 256, 1024, 4096, 16384 and the complete thirteen-step word below. Stored-body Hamiltonians are zero in this model. Physical promotion is zero.

II. POSITIVE FILTER-GRAM GENERATOR

Finite-time coarse graining can preserve positivity beyond a fully secular approximation [1]. Our explicit centered-window coefficient matrix is

$$K_{ab}^{(\tau)} = \frac{\tau}{2\pi} \int_{-\infty}^{\infty} \gamma(\omega) f_a(\omega) f_b(\omega) d\omega, \quad (1)$$

$$f_a(\omega) = \text{sinc}[(\omega - \nu_a)\tau/2], \quad (2)$$

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where $\text{sinc } x = \sin x/x$. The signed dressed frequencies ν_a include lowering and raising operators and their cross terms. For $\omega > 0$,

$$\gamma(\omega) = \frac{\alpha\omega e^{-\omega/640}}{1 - e^{-\beta\omega}}, \quad \gamma(-\omega) = e^{-\beta\omega} \gamma(\omega), \quad (3)$$

with α chosen so $\gamma(64) = 8$. The coefficient matrix gives the dissipator

$$\sum_{ab} K_{ab} \left[A_a \rho A_b^\dagger - \frac{1}{2} \{ A_b^\dagger A_a, \rho \} \right]. \quad (4)$$

For every complex vector z , $z^\dagger K z$ is an integral of $\gamma(\omega) |\sum_a z_a f_a(\omega)|^2$ times a positive factor. Therefore $K \succeq 0$ exactly. Positive quadrature also preserves the Gram form. Positivity alone does not establish approximation accuracy, thermal detailed balance at finite τ or hardware suitability.

The generator uses the laboratory Hamiltonian. Raising/lowering cross terms need not preserve the free-energy block structure, so replacing the laboratory Hamiltonian by the interaction Hamiltonian would alter this workload. Centered static semigroups are the declared construction; they are not asserted to be an exact reduced microscopic evolution.

III. NUMERICAL FILTERING AND COMPARISON DEBIT

Use a signed spectral cutoff of $48\omega_c$ and composite positive Gauss–Legendre rules of orders 16 and 32. An initial panel-width rule $4/\tau$ failed the coefficient-convergence criterion near the thermal spectral scale. A recorded numerical repair limits width to $\min(4/\tau, 1)$; spectrum, coarse-graining choices and tolerance remain unchanged. The repaired rules agree within $1.6 \cdot 10^{-14}$ in maximum coefficient entry. Tiny negative floating eigenvalues, at most about $2.8 \cdot 10^{-14}$ in magnitude, are treated as rounding in jump construction. These are numerical controls, not interval-certified quadrature error bounds.

Let K_0 represent the original local jump: all lowering-block entries equal eight and all other entries zero. For two specified GKSL generators with the same Hamiltonian, CPTP Duhamel contraction gives

$$\frac{1}{2} \|\Phi_t^{(K)} - \Phi_t^{(K_0)}\|_\diamond \leq \min\{1, tB\}, \quad (5)$$

$$B = \sum_{ab} |K_{ab} - K_{0,ab}| \|A_a\| \|A_b\|. \quad (6)$$

The bilinear dissipator norm is at most $2\|A_a\| \|A_b\|$, which establishes the inequality. It bounds a comparison between chosen generators, not error relative to the bath. Computed coefficients are not interval-enclosed. Every tested raw debit exceeds one, so this comparison provides no small-error

TABLE I
FROZEN COARSE-GRAINED PORT OUTCOMES AT $t = 16$

τ	Retained target error	Difference from local model
1/64	0.689243	0.689239
1/16	0.307529	0.307525
1/4	0.108757	0.108753
1	0.039583	0.039579
4	0.006285	0.006281

certificate. All five fail 2^{-16} numerically. Finite filter width admits contributions to the raising block from positive-frequency spectral weight; those coefficients must not be replaced by the single reverse ratio at Ω . These failures are scoped to this frozen filtered model, not a universal reservoir no-go result.

An analytic tail envelope is derived separately. The inequalities $\alpha < 5/36$, $\pi > 3$ and $e^{-48} < 2^{-69}$ bound the omitted spectrum. The last inequality is verified by a positive degree-100 rational Taylor lower sum for e^{48} . With at most 56 signed frequencies for eight nondegenerate levels, $\|A_a\| \leq 1$, $\tau \leq 4$ and $t \leq 17$, this bounds the omitted-tail comparison debit. It does not bound interior quadrature, Born, memory or Lamb-shift errors. Those microscopic debits remain OPEN.

IV. COMPLETE BANK WORD AND HISTORY GAUGE

The body has nine qubits: $A, J, O_1, O_2, D_{1A}, D_{1J}, D_{2A}, D_{2J}, E$. The body exhaustion qubit E is distinct from the fresh completion flag F . The exact word is

$$\begin{aligned} &H_A, T_A, \text{CNOT}_{A,J}, \text{SWAP}_{A,O_1}, \\ &\text{SWAP}_{A,D_{1A}}, \text{SWAP}_{J,D_{1J}}, \\ &T_A, H_A, \text{CNOT}_{A,J}, \text{SWAP}_{A,O_2}, \\ &\text{SWAP}_{A,D_{2A}}, \text{SWAP}_{J,D_{2J}}, X_E. \end{aligned}$$

The clock has addresses $0, \dots, 13$, with $p = 13$, plus terminal states m, e, d : 17 native levels. Launch is address zero with $F = R = 0$. The same static terminal marker flips F only on $p-m$ traversal, followed by flag-conditioned $m-e$ entry and the auxiliary exchange.

Let U_j be gate j , $P_0 = I$ and $P_j = U_j \cdots U_1$. Assign P_{13} to each terminal state m, e, d , and define

$$V = \sum_{j=0}^{16} |j\rangle\langle j| \otimes P_j \otimes I_{FR}. \quad (7)$$

The labelled Hamiltonian is $V(K_{CFR} \otimes I_B)V^\dagger$. Auxiliary bath operators commute with V . This exact history gauge reduces the cold dynamics to a scalar live chain while every captured branch carries the complete unitary P_{13} . All donor/output registers and arbitrary reference spectators are included; none are reset by tracing them out.

For fixed launch, the cold state is a surviving live block plus a dark captured block. These supports are orthogonal, so retained half-diamond distance to the complete-word target equals survival $q(t)$. This argument applies to arbitrary body/reference inputs. The all-zero fixture is merely a numerical check: its captured pairs (O_1, D_{1J}) and (O_2, D_{2J}) both have negativity $1/2$, with A, J fresh and $E = 1$.

TABLE II
COMPLETE-WORD COLD SCALAR SURVIVAL; LOCAL MODEL ONLY

Time	Retained channel error
16	0.114917
64	0.013971
256	0.001603
1024	$7.339677 \cdot 10^{-6}$
4096	$3.757924 \cdot 10^{-15}$
16384	$2.615117 \cdot 10^{-52}$

V. ARRIVAL CERTIFICATE AND RESOURCE BOUNDARY

The live chain has 17 vertices, uniform weights except final exchange λ , and terminal damping four in its no-jump generator A . Exact algebra gives $A + A^T = -8P_{\text{last}}$, hence $q'(t) = -8|a_{\text{last}}(t)|^2 \leq 0$. The first passing frozen sample, 1024, was selected for independent interval verification; it is not a predicted or minimum deadline. At the exact left edge $1048575/1024$, an 80-digit directed interval exponential, Taylor order 24 and scaling 2^{20} yields

$$q < \frac{1}{131072} < \frac{1}{65536}. \quad (8)$$

Monotonic survival extends the cold certificate across $|t - 1024| \leq 1/1024$ and later cold times. For the local reverse jump, the sufficient range $r \leq 2^{-31}$ gives the uniform bound

$$\begin{aligned} \delta(t) &\leq 2^{-17} + 8 \cdot 2^{-31}(1024 + 1/1024) \\ &= \frac{3145729}{274877906944} < 2^{-16}. \end{aligned} \quad (9)$$

The longer episode tightens the conservative reverse-rate budget. This bound does not apply to the failed coarse-grained Ohmic models. The native full dimension is $17 \cdot 2 \cdot 2 \cdot 512 = 34816$, and the admitted dimension is 17408. A five-bit clock embedding is only an address bound; it is not a physical qubit two-local compiler. History edges for CNOT and SWAP generally act on a clock and two body qubits. Native joint admission, geometry and a matched physical gate compiler remain OPEN. A full scalar-spectrum norm audit charges all clock/flag/auxiliary sectors; it does not use the smaller live path to hide costs.

The replay contains 21 exact word checks, 46 numerical bath/word checks, the directed interval certificate and frozen failures. No hardware result, quantum advantage, microscopic bath certificate or full physical architecture promotion follows. Physical promotion remains zero. Next work should test an explicitly energy-conserving reservoir coupling under a fresh hypothesis, while benchmarking arrival profiles under a matched full-spectrum budget. The present failed bath family and slower word baseline must remain frozen.

REFERENCES

- [1] G. Schaller and T. Brandes, "Preservation of positivity by dynamical coarse graining," *Phys. Rev. A*, vol.78, 022106, 2008. Author preprint: arXiv:0804.2374v2.